



Metadata standards and controlled vocabularies

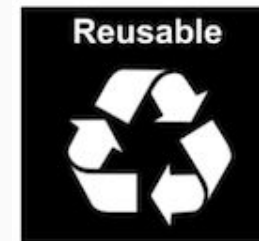
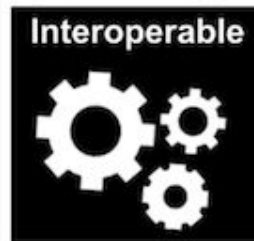
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elixir.no

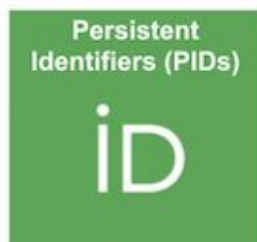
What is a metadata standard

- How does a standard make data FAIR?
- What are the ingredients for defining a standard?
- Where to find metadata standards?
- Which tools can be used in connection with standards?



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Metadata standards and “Findable”, “Accessible”

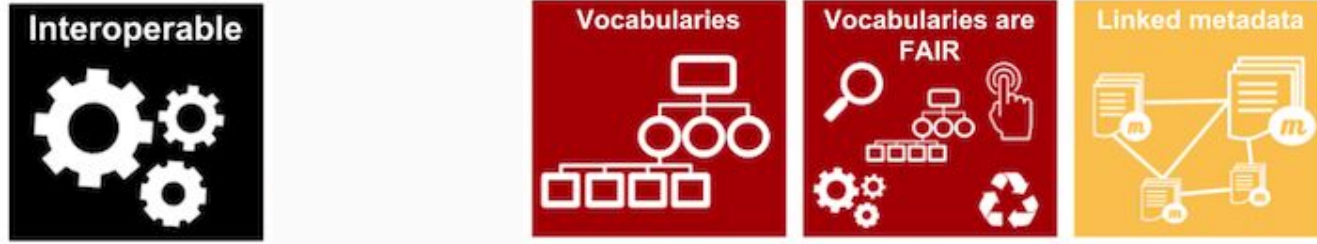


- The metadata requires a persistent identifier
- Metadata standard compliant with repositories checklist
- Metadata fields used to identify/retrieve the data



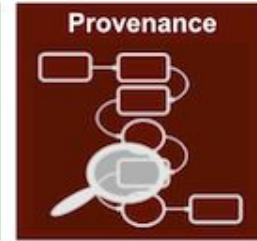
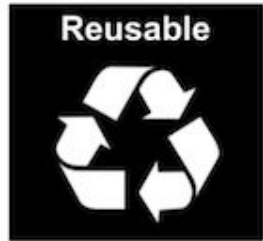
- Persistency of metadata (also when the data is not available)

Metadata standards and “Interoperable”



- Interoperability fully relies on metadata
- Vocabularies and ontologies ensure that standards are FAIR
- A well-defined standard can be linked with standards describing other type of data
- Focus on machine-actionability

Metadata standards and “Reusable”



- Richer metadata fields enhance reusability
- Metadata should describe data provenance
- Everything needs to follow community standards
 - Alignment with repository



MINSEQE; MIAME



HUPO-PSI ; TraML; MIAPE



SRA-XML

Example: ENA sample checklists 1



There is a minimum amount of information required during ENA sample registration and all samples must conform to a defined checklist of expected metadata values. The most suitable checklist depends on sample type.

<https://www.ebi.ac.uk/ena/browser/checklists>

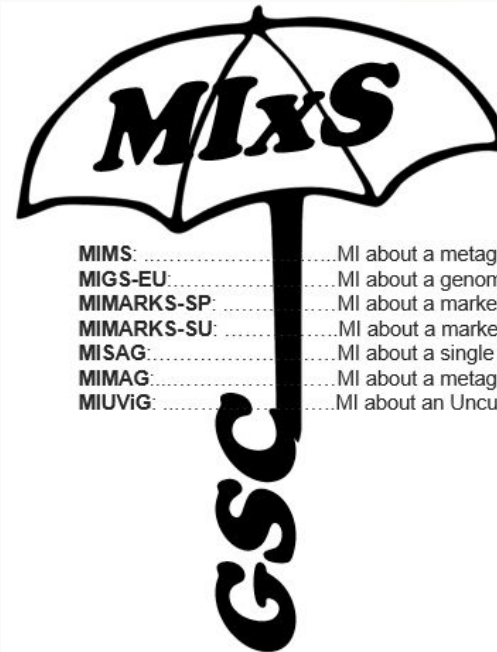
Example: ENA sample checklists 2



Sample Checklists

Accession	Name
ERC000012	GSC MixS air
ERC000013	GSC MixS host associated
ERC000014	GSC MixS human associated

<https://www.ebi.ac.uk/ena/browser/checklists>



MIMS:MI about a metagenome sequence.
MIGS-EU:MI about a genome sequence
MIMARKS-SP:MI about a marker gene sequence obtained directly from the environment
MIMARKS-SU:MI about a marker gene sequence from cultured or voucher-identifiable specimens
MISAG:MI about a single amplified genome sequence.
MIMAG:MI about a metagenome-assembled genome sequence.
MIUViG:MI about an Uncultivated Virus Genome

<https://www.gensc.org/pages/standards-intro.html>

Example: ENA sample checklists 3



ERC000014 GSC MlxS human
associated

Field Name	Field Format (Field Restriction)	
project name	?	free text
experimental factor	?	free text
ploidy	?	free text
number of replicons	?	restricted text regular expression ?

<https://www.ebi.ac.uk/ena/browser/checklists>

Example: ENA sample checklists 4



ERC000014

GSC MlxS human
associated

number of replicons



restricted text

regular expression ?

smoker



text choice

ex-smoker

non-smoker

smoker

<https://www.ebi.ac.uk/ena/browser/view/ERC000014>

Controlled vocabularies

Lists of pre-selected answers for metadata fields are known as **controlled vocabularies**. They are useful to avoid situations such the one below, where things got out of control

How many ways can you say “female”?

18-day pregnant females	female (lactating)	individual female	worker caste (female)
2 yr old female	female (pregnant)	lgb*cc females	sex: female
400 yr. old female	female (outbred)	mare	female, other
adult female	female parent	female (worker)	female child
asexual female	female plant	monosex female	femal
castrate female	female with eggs	ovigerous female	3 female
cf.female	female worker	oviparous sexual females	female (phenotype)
cystocarpic female	female, 6-8 weeks old	worker bee	female mice
dikaryon	female, virgin	female enriched	female, spayed
dioecious female	female, worker	pseudohermaprhoditic female	femlale
diploid female	female(gynoecious)	remale	metafemale

Controlled vocabularies – when not used

How many ways can you say “female”?

18-day pregnant females	female (lactating)	individual female	worker caste (female)
2 yr old female	female (pregnant)	lgb*cc females	sex: female
400 yr. old female	female (outbred)	mare	female, other
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castrate female	female with eggs	ovigerous female	3 female
cf.female	female worker	oviparous sexual females	female (phenotype)
cystocarpic female	female, 6-8 weeks old	worker bee	female mice
dikaryon	female, virgin	female enriched	female, spayed
dioecious female	female, worker	pseudohermaphroditic female	femlale
diploid female	female(gynoeceious)	remale	metafemale
f	femele	semi-engorged female	sterile female
famale	female, pooled	sexual oviparous female	normal female
femal	femalen	sterile female worker	sf
female	females	strictly female	vitellogenic replete female
female - worker	females only	tetraploid female	worker
female (alate sexual)	gynoeceious	thelytoky	hexaploid female
female (calf)	healthy female	female (gynoeceious)	female (f-o)
hen	probably female (based on morphology)		

female (note: this sample was originally provided as a \"male\" sample to us and therefore labeled this way in the brawand et al. paper and original geo submission; however, detailed data analyses carried out in the meantime clearly show that this sample stems from a female individual)

Courtesy of N. Silvester, European Nucleotide Archive, EMBL-EBI

Ontologies 1

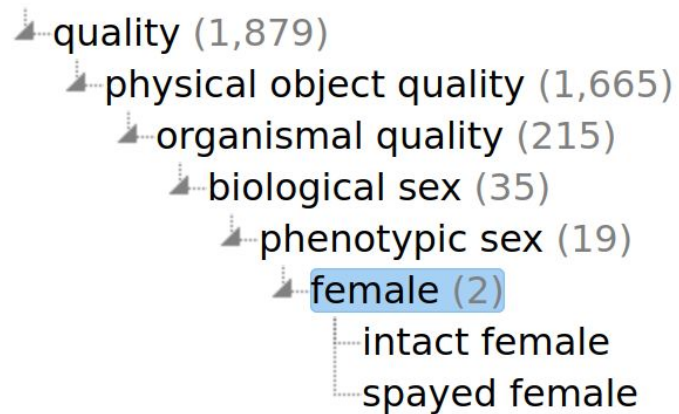
While **controlled vocabularies** offer more control over free text boxes, this solution is not ideal as different data sources could use different options with e.g. a different granularity.

The way to address this is to replace words in this list with **persistent identifiers** provided by an external sources. This would enable cross-linking between data resources i.e. **interoperability**



Ontologies 2

Granularity is very important to find the right terminology. Using a well-defined hierarchy usually helps finding the right information.



Phenotype And Trait Ontology

An ontology of phenotypic qualities (properties, attributes or characteristics)

PID

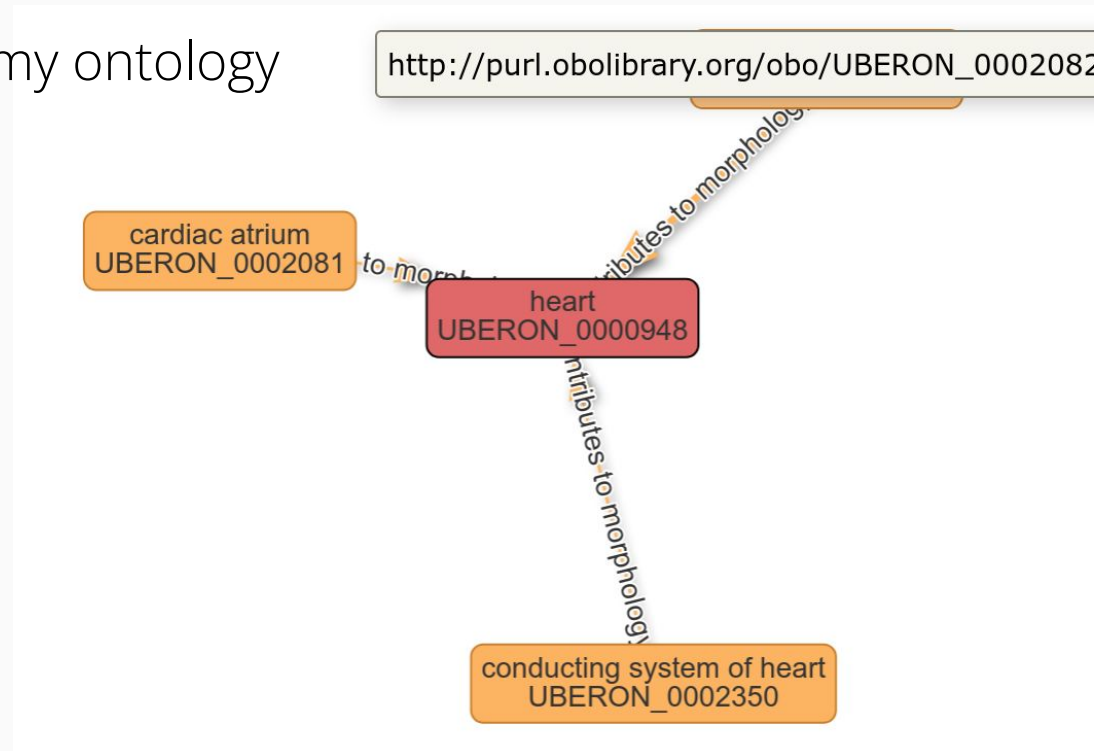
http://purl.obolibrary.org/obo/PATO_0000383

https://www.ebi.ac.uk/ols4/ontologies/pato/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FPATO_0000383

Ontologies 3

- On top of the **hierarchy**, formal relations between terms are also relevant to interlink data
- These can be visualised as **graphs**

Uber-anatomy ontology



surrounds	☐
in right side of	☐
in left side of	☐
contributes to morphology of	☒
has part	☐
continuous with	☐
channels_from	☐
results in formation of	☐
results in growth of	☐
has soma location	☐
develops from	☐

Ontologies 4

Cross-linking of terms across ontologies allows for an improved standardisation of terms and definitions.

heart

http://purl.obolibrary.org/obo/UBERON_0000948  Copy

Also appears in

OVAE

OPMI

CPONT

ZP

AIMS

EFO

CLO

OBIB

MONDO

TXPO

OHPI

XPO

FOVT

PCL

ENVO

WBPHENOTYPE

OBI

RBO

NBO

HP

GAZ

OBA

ONE

ECTO

EUPATH

HCAO

FLOPO

GENEPIO

ICO

FOODON

BCGO

CCF

OAE

MICRO

MAXO

MP

ECOCORE

GO

MS

CL

DOID

PATO

https://www.ebi.ac.uk/ols4/ontologies/uberon/classes/http%253A%252F%252Fpurl.obolibrary.org%252Fobo%252FUBERON_0000948

Ontologies 5

Definition and **synonyms** to facilitate search functionalities.

heart

http://purl.obolibrary.org/obo/UBERON_0000948  Copy

A myogenic muscular circulatory organ found in the vertebrate cardiovascular system composed of chambers of cardiac muscle. It is the primary circulatory organ. 

Synonym

branchial heart 

cardium 

chambered heart 

vertebrate heart

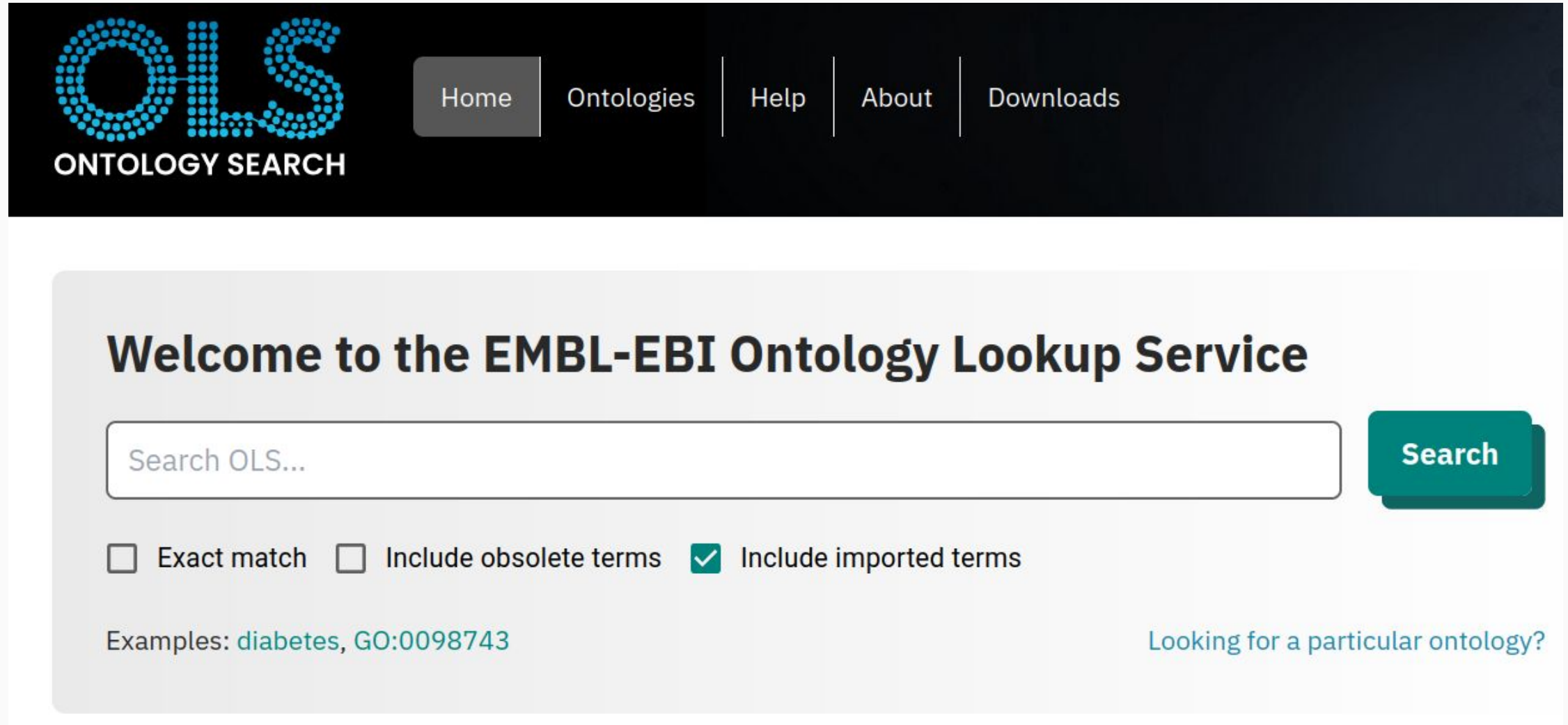
Ontologies – summary

To summarise, an ontology is a set of concept and categories in a subject area that:

- Shows and define properties
- Shows the relation between these properties
- Provides a persistent identifier to these properties
- Follows a specific tree-like hierarchy
- Is cross-linked with other ontologies



Ontologies – where to find them 1



The screenshot shows the EMBL-EBI Ontology Lookup Service (OLS) homepage. At the top is a dark blue navigation bar with the OLS logo (a stylized 'OLS' made of dots) and the text 'ONTOLOGY SEARCH' on the left. To the right of the logo are five navigation links: 'Home' (highlighted with a grey background), 'Ontologies', 'Help', 'About', and 'Downloads'. Below the navigation bar is a large light grey box containing the main content. The heading 'Welcome to the EMBL-EBI Ontology Lookup Service' is centered at the top of this box. Below the heading is a search interface consisting of a text input field with the placeholder 'Search OLS...' and a teal 'Search' button to its right. Under the input field are three checkboxes: 'Exact match' (unchecked), 'Include obsolete terms' (unchecked), and 'Include imported terms' (checked). Below the checkboxes, the text 'Examples: diabetes, GO:0098743' is displayed. On the right side of the light grey box, the text 'Looking for a particular ontology?' is visible.

OLS
ONTOLOGY SEARCH

Home | Ontologies | Help | About | Downloads

Welcome to the EMBL-EBI Ontology Lookup Service

Search OLS...

☐ Exact match ☐ Include obsolete terms ☒ Include imported terms

Examples: [diabetes](#), [GO:0098743](#)

[Looking for a particular ontology?](#)

<https://www.ebi.ac.uk/ols4/>

Ontologies – where to find them 2

Welcome to BioPortal, the world's most comprehensive repository of biomedical ontologies

Search for a class

Enter a class, e.g. Melanoma 

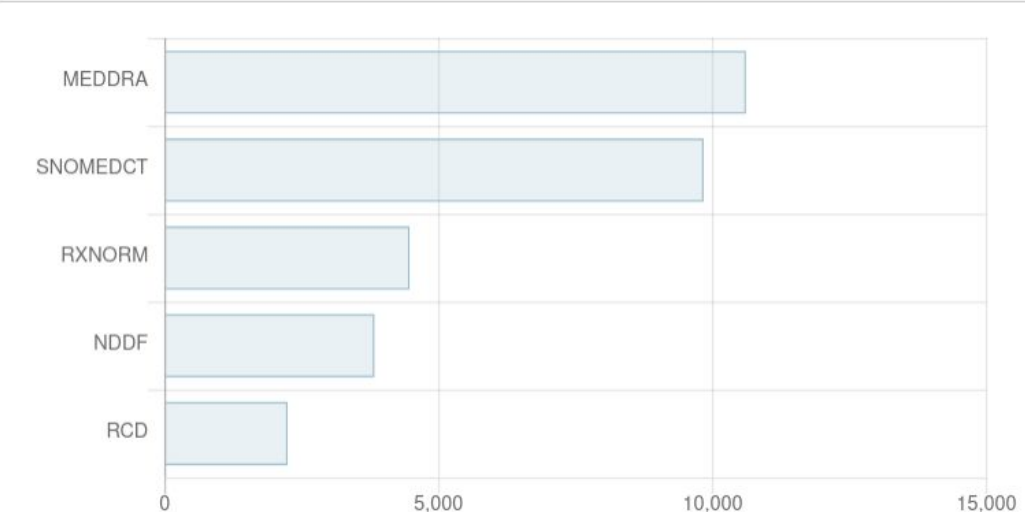
[Advanced search](#)

Find an ontology

Start typing ontology name, then choose from list 

[Browse ontologies](#)

Ontology visits (February 2025)



Ontology	Visits (approx.)
MEDDRA	11,000
SNOMEDCT	10,000
RXNORM	4,500
NDDF	4,000
RCD	2,500

Statistics

Ontologies	1,185
Classes	15,335,435
Properties	36,286
Mappings	102,816,942

<https://bioportal.bioontology.org/>



Ontologies - where to find them 3

FAIRsharing.org standards, databases, policies

search through all content

SEARCH ADVANCED SEARCH LOGIN

STANDARDS DATABASES POLICIES COLLECTIONS ORGANISATIONS ADD CONTENT STATS

A curated, informative and educational resource on data and metadata standards, inter-related to databases and data policies.

1861 Standards

Terminology Artifact	812
Model/Format	641
Reporting Guideline	317
Identifier Schema	68

VIEW ALL

2322 Databases

Repositories	1222
Knowledgebases	927
Knowledgebase/Repositories	173
Institutional Repositories	96

VIEW ALL

344 Policies

Journal	193
Institution	53
Funder	40
Journal publisher	35
Society	17
Project	6

VIEW ALL

Not for browsing ontologies, but to discover relations between ontologies, standards and repositories

[https://fairsharing.org/search?recordType=terminology artefact](https://fairsharing.org/search?recordType=terminology+artefact)

Metadata tracking platforms

Domain specific:

- COPO for plant sciences
- MOLGENIS for biobanking
- Omero for imaging data



MOLGENIS



Customisable (domain expertise required)

- Proprietary ELNs/LIMS -
 - often poor support for ontologies
- openBIS - open source ELN/LIMS
- FAIRDOM SEEK



<https://copo-project.org/>

<https://www.molgenis.org/>

<https://openbis.ch/>

<https://seek4science.org/>

 [Browse](#) · [Create](#) · [Help](#) ·  Natalie Stanford

Sample type was successfully created. 

Editing Sample Type

Title *

Description

baking an apple pie

Projects *

The following projects are associated with this sample type:

Default Project [\[remove\]](#)

Select Project ...

Tags

Attributes

Re-arrange attributes by clicking and dragging the button on the left-hand side of each row.

Order	Name	Required?	Title?	Type	Unit	
  1	Bake ID	☒	☒	String		Remove
  2	Date of bake	☐	☐	String		Remove
  3	Cooking temperature	☐	☐	String		Remove
  4	Cooking time	☐	☐	String		Remove
  5	Type of Apple	☐	☐	String		Remove

FAIRDOM SEEK

The [SEEK](#) platform is a web-based resource for sharing heterogeneous scientific research datasets, models or simulations, processes and research outcomes. It preserves associations between them, along with information about the people and organisations involved.

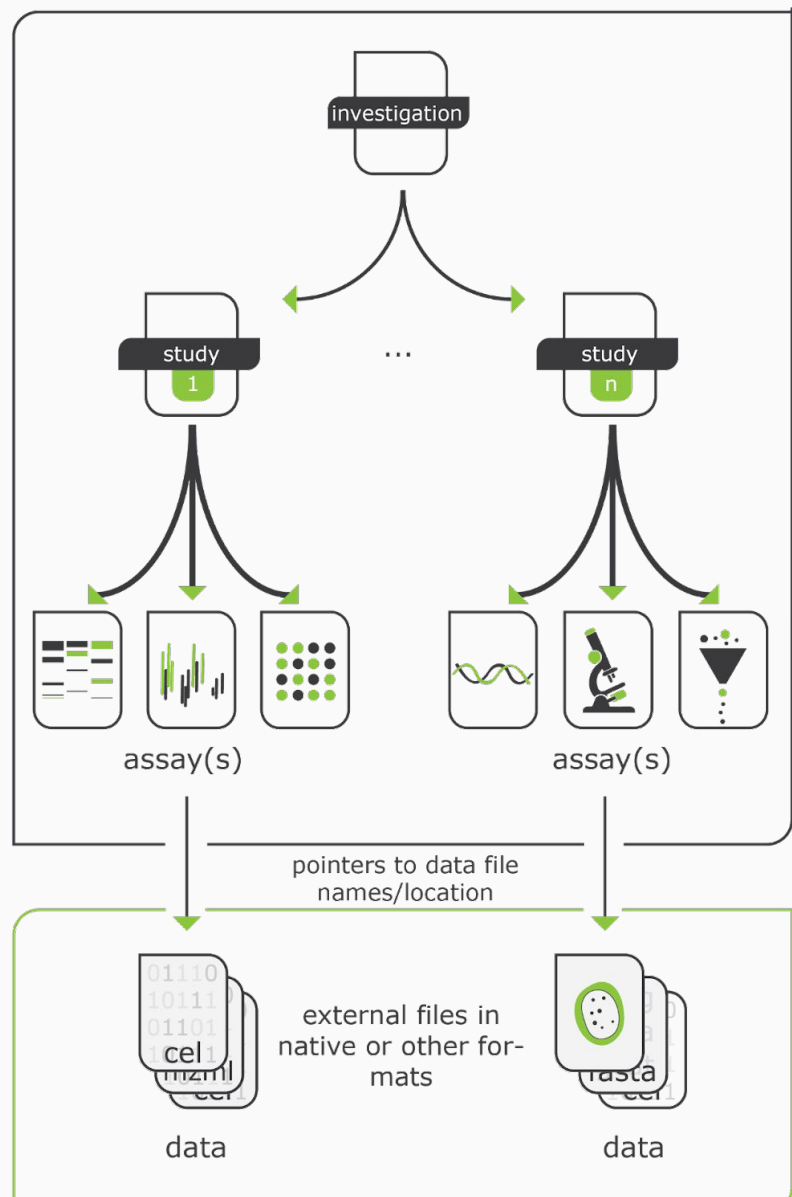
- National users (via Digital Life):



- Target new users (via ELIXIR)



The ISA model



- Investigation
 - Persons
 - Organizations
 - Publications
- Study(s)
 - Design
 - Factor
 - Protocol
- Assay(s)
 - Measurement
 - Technology
 - Materials
 - Data

Domain-agnostic standard with typical metadata fields for a project in the life sciences

Possibility of semantic annotations



Standards and repositories - where to find them



search through all content

SEARCH

LOGIN

STANDARDS

DATABASES

POLICIES

COLLECTIONS

ORGANISATIONS

ADD CONTENT

STATS

A curated, informative and educational resource on data and metadata standards, inter-related to databases and data policies.

Guides consumers to discover, select and use these resources with confidence.

Helps producers to make their resources more visible, more widely adopted and cited.

Provides humans and tools with access to trustworthy content to enable data management tasks.



Thank you



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